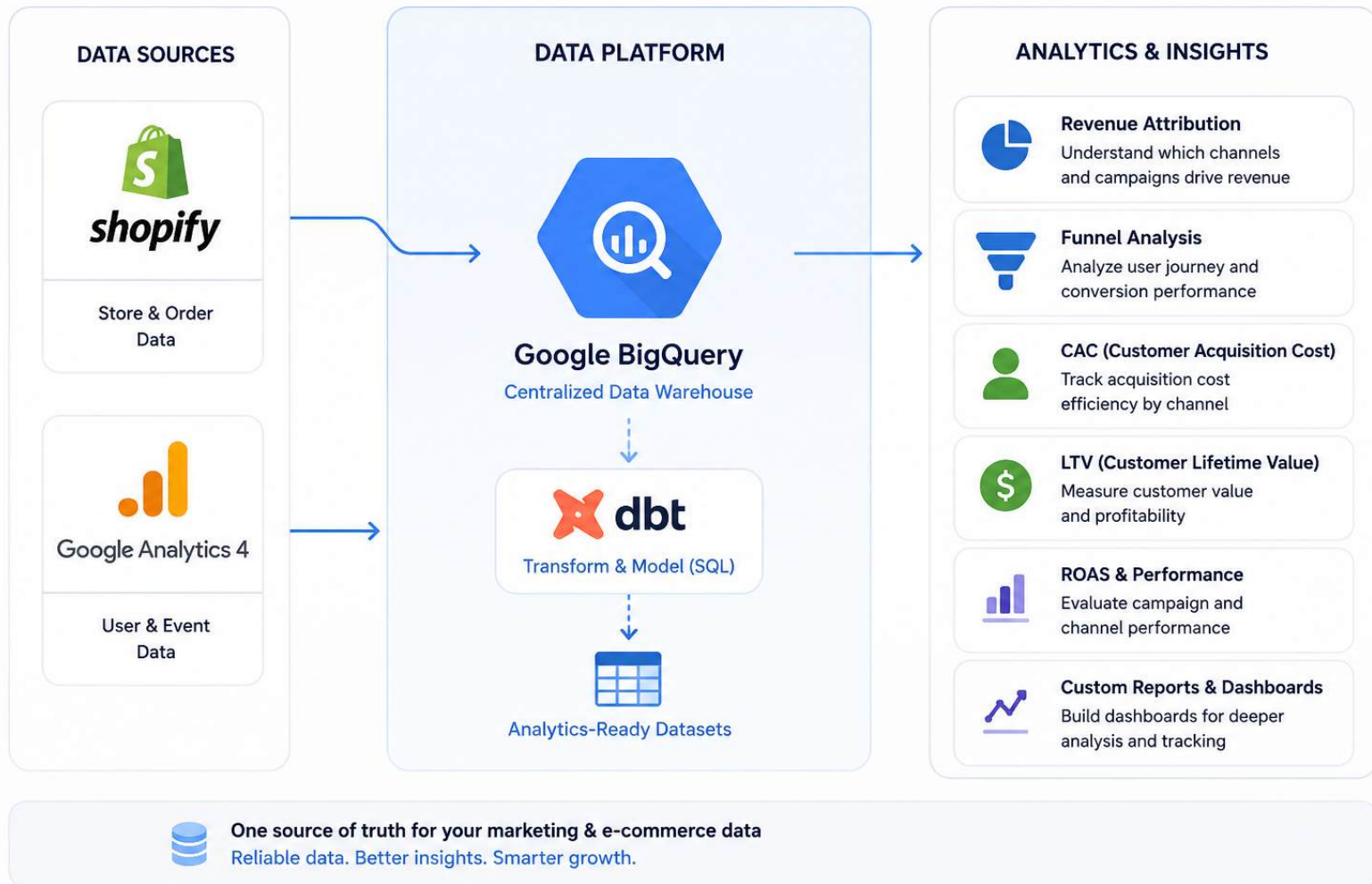




Shopify + GA4 Full-Funnel Growth & Revenue Attribution Engine

Connecting ecommerce behavior, orders, revenue, and attribution from raw data to dashboard-ready insights.





The Business Problem

Shopify knows **what** was purchased — but not which sessions or channels influenced it. GA4 knows **how** users behaved — but its purchase data needs reconciliation before it can be trusted.

Revenue Attribution Gap

Which channels are associated with real Shopify revenue?

Funnel Visibility

Where do users drop off before completing a purchase?

Attribution Coverage

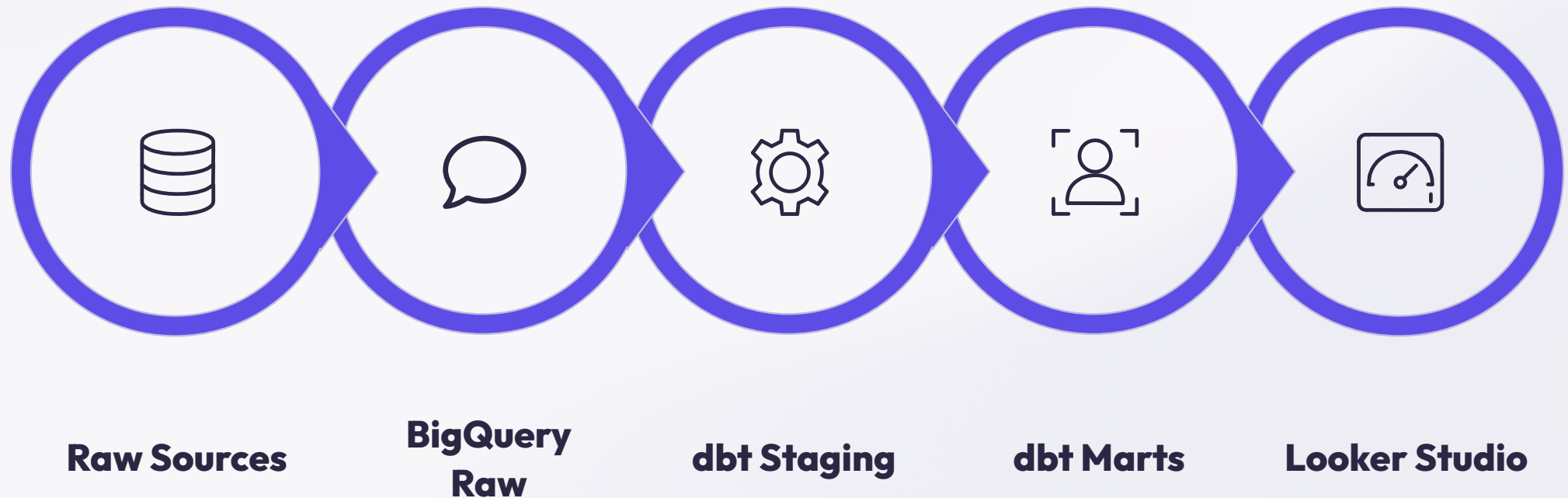
How much revenue is attributed vs. left in the unattributed bucket?

Data Trust

Can GA4 purchase events be reconciled against Shopify order truth?

Solution Overview

An end-to-end analytics pipeline connecting raw ecommerce sources to a four-page Looker Studio dashboard — following the full arc from **traffic** → **behavior** → **conversion** → **revenue** → **attribution**.



Shopify is the source of truth for orders and revenue. GA4 provides behavioral context, session attribution dimensions, and funnel event data — unified through a reconciliation layer before attribution is applied.

Data Ingestion Architecture

Shopify Ingestion (Python)

Custom Python scripts call the **Shopify Admin API** to load three raw tables into BigQuery:

Incremental loads use **updated_at** timestamps and a **pipeline_state** checkpoint table. Raw JSON payloads are retained intact for full auditability.

raw_shopify_orders

raw_shopify_products

raw_shopify_customers

analytics_534648282.events_YYYYMMDD

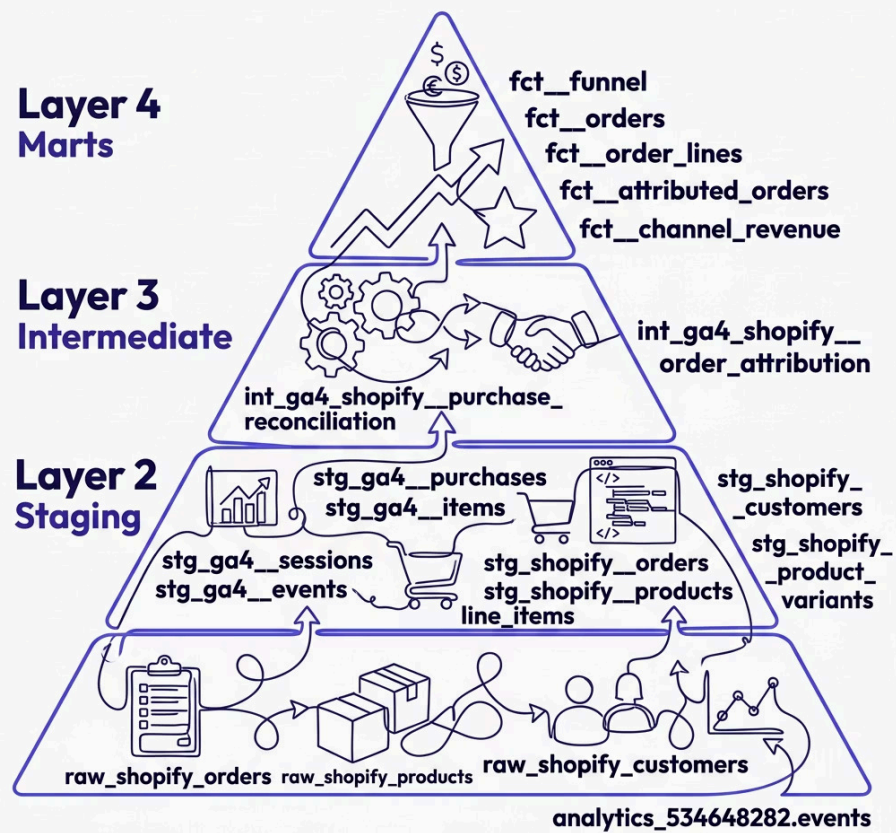
One row per GA4 event, partitioned by date. Nested JSON fields are unnested in dbt staging.

GA4 Ingestion (Native Export)

GA4 uses BigQuery's native daily export — no custom ingestion required. Tables land as:

 Retaining raw JSON ensures any modeling decision can be re-derived without re-pulling from source APIs.

dbt Modeling Architecture



Key Model Grains

Events

One row per GA4 event

Sessions

One row per GA4 session

Reconciliation

One row per GA4 purchase event — the audit proof layer

Attribution

One row per Shopify order

Channel Revenue

One row per date + currency + channel/campaign

Reconciliation Logic

GA4 purchase events must be validated against Shopify orders *before* attribution is applied. The match rule is simple and auditable:

 **Match Rule:** GA4 `transaction_id` = `CAST(Shopify order_id AS STRING)`

15

GA4 Purchases

Total purchase events in current dataset

12

Matched

Confirmed via order_id match

3

Unmatched

1 null transaction_id, 2 with value "not_set"

-  Unmatched GA4 purchases are **retained** in the reconciliation layer — it is the audit/proof layer, not the attribution layer. Unmatched Shopify orders are preserved in an **unattributed bucket** to ensure revenue totals always reconcile.

Attribution & Revenue Modeling



Design Principles

→ **Shopify = Revenue Truth**

Order totals, dates, and IDs always come from Shopify — never from GA4.

→ **GA4 = Attribution Context**

Source, medium, and campaign are attached only where a confirmed match exists.

→ **No Revenue Left Behind**

Unmatched orders remain in the unattributed bucket, ensuring 100% revenue coverage.

Looker Studio Dashboard Experience

Four dashboard pages built entirely from dbt marts — each designed to answer a distinct stakeholder question.



Executive Overview

What happened? — Top-line revenue, order volume, and channel summary.



Funnel Performance

Where did users move or drop off? — Session-to-purchase conversion by stage.



Channel Attribution

Which channels drive revenue? — Attributed vs. unattributed by source/medium/campaign.



Reconciliation & QA

How trustworthy is the matching? — Match rates, unmatched records, and data quality signals.



Data Quality, Testing & Known Limitations

dbt Testing Layer

Custom tests validate engineering assumptions at every layer:

GA4 Purchase Integrity

Assert transaction_id is not null

Shopify Order Totals

Assert non-negative revenue values

Order Line Quantities

Assert positive unit quantities

Reconciliation Audit

Flag unmatched GA4 purchases as known signals

Revenue Diff Checks

Alert on large GA4 vs. Shopify revenue discrepancies

Known Data Limitation



The current dataset consists primarily of **setup, debug, and manual test traffic**. Sources include tagassistant, admin.shopify, and referral — with many records carrying weak source/medium/campaign values.

This is expected and intentional. The project validates:

- Pipeline architecture and ingestion logic
- dbt modeling and reconciliation correctness
- Attribution logic and dashboard structure

Funnel and channel outputs are **validation results**, not production marketing conclusions.



Outcomes & Next Steps

What Was Delivered

01

End-to-End Pipeline

Shopify Admin API + GA4 export →
BigQuery → dbt → Looker Studio

02

Documented Marts

Raw nested JSON modeled into 5
production-ready mart models

03

Reconciliation Layer

GA4 purchases validated against
Shopify order truth at purchase-
event grain

04

Attribution Facts

Session-based attribution with full
unattributed revenue coverage

Roadmap

→ Production Traffic

Real UTM standards and
internal traffic exclusion

→ Source Normalization

Standardize
source/medium/campaign
values

→ Orchestration

Dagster pipeline
orchestration for scheduled
runs

→ Future Scope

CAC, LTV, and profitability
modeling project

This project demonstrates **analytics engineering depth** — data modeling, reconciliation thinking, attribution strategy, and BI storytelling — across the full ecommerce growth funnel.